

Ritam Mukherjee

Columbus, OH • mukherjee.241@osu.edu • +1-614-405-5118 • [LinkedIn://ritammukherjee27](https://www.linkedin.com/in/ritammukherjee27) • [Github://RitamCODE](https://github.com/RitamCODE) • ritam-stallion27.netlify.app

EDUCATION

THE OHIO STATE UNIVERSITY

Columbus, OH

M.S. Computer Science and Engineering

Aug 2024 -- May 2026

GPA: 3.4/4.0. Coursework: Algorithms, Neural Networks, AI, Speech and Language Processing, Compiler Design.

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Bhubaneswar, India

B.Tech Computer Science and Engineering

Sept 2020 -- May 2024

GPA: 3.7/4.0. Coursework: DSA, Design and Analysis of Algorithms, Cloud Computing (AWS EC2, S3, Lambda), DBMS, High Performance Computing.

EXPERIENCE

OHIO STATE UNIVERSITY

Columbus, OH

Graduate Researcher -- AO-OCT Retinal Imaging

2025 -- Present

- Extended a published PyTorch and HuggingFace (SegFormer-B5) retinal segmentation pipeline to export raw logits and compute three confidence metrics, max softmax probability, entropy, and margin, for every inference pass, with no retraining or additional compute.
- Designed and implemented a strip-based spatial confidence-analysis framework in Python, NumPy, and Matplotlib, run in batch across 47 test frames on the Ohio Supercomputer Center's SLURM/HPC cluster with results tracked in Weights & Biases; segmentation F1 dropped from 0.9443 to 0.6597 under domain shift, with confidence disturbances spatially aligned to disease pathology and no disease-specific labels required.

Advisor: Dr. Rajiv Ramnath

OHIO STATE UNIVERSITY

Columbus, OH

Teaching Assistant -- Full-Stack Dev

Aug 2025 -- Apr 2026

- Held weekly office hours and provided debugging support for students building and deploying full-stack web applications and REST APIs in JavaScript, TypeScript, React, .NET, and C#; graded assignments and projects across the course.

THEDELTACUBE.AI

Remote, India

Junior Data Scientist

Feb 2024 -- May 2024

- Served as primary developer of Brainbox, an AI document-intelligence platform for multilingual newspaper agencies, covering summarization, Q&A, translation, and image generation.
- Scaled the summarization pipeline from 5–10 pages to 400–500 pages by migrating load_qa to RetrievalQA and adding a chunking plus vector-store retrieval layer, eliminating token-overflow failures.
- Benchmarked 10+ closed- and open-source LLMs (GPT-3.5, GPT-4, Llama 2, Gemma, CodeLlama) and evaluated FAISS against ChromaDB, selecting Gemma 2B and FAISS for the best cost-to-performance and runtime tradeoff.
- Refactored a monolithic codebase into a modular architecture with structured exception handling, and shipped UI improvements to the Streamlit front end.

MICROSOFT

Bangalore, India

Software Engineer Intern

May 2023 -- Jul 2023

- Selected for the internship out of an all-India Microsoft hackathon field on the strength of an independently built facial-recognition credential-storage web app, developed before Windows Hello existed.
- Independently designed and implemented a serialization fix for LangChain RetrievalQA chains in MLflow (25.5k+ GitHub stars) and submitted it as a pull request; an external contributor's functionally identical solution was merged into the same repo the same day, validating the approach.
- Added an elif branch to MLflow's _save_model module and a new rdocs parameter to persist retriever state as a pickle file, preserving full RAG functionality across the save and load lifecycle.
- Integrated tracing, logging, and memory management into AzureML Gen-AI workflows, and delivered a standardized path for deploying LangChain RAG applications through the AzureML UI, removing manual packaging overhead for engineers.

PROJECTS

ADAPTMATH -- K-5 ADAPTIVE MATH TUTOR, NERDY AI HACKATHON

Python, FastAPI, React/Vite, LangGraph, Bayesian Knowledge Tracing

- Built a full-stack K–5 adaptive math tutoring app (React/Vite, FastAPI) combining Bayesian Knowledge Tracing with a LangGraph state machine to route each student to the next problem, a retry, remediation, or a skill demotion, keeping grading and curriculum logic fully deterministic and independent of LLM output.
- Designed 13 rule-based misconception detectors that diagnose the specific error behind a wrong answer (e.g. summing columns instead of carrying) rather than just marking it incorrect, backed by 100+ pytest unit tests.
- Restricted all LLM calls (OpenAI API) to 4 isolated narration touchpoints, holding grading latency to 6–15ms against a 150ms budget with zero LLM calls on the grading path.
- Built a synthetic-student evaluation harness (4 ability profiles, n=200/condition) using two-proportion z-tests with BH-FDR correction to test whether the adaptive engine improved learning over a baseline, and reported the result honestly rather than overstating it.

COMPARING SMALL LLMS: FINE-TUNED QA VS. FULL-CONTEXT

Python, PyTorch, HuggingFace Transformers, LoRA

- Fine-tuned Qwen2.5-0.5B with LoRA under two strategies, QA-pair tuning versus full-text context, and built an automated blind-judge evaluation pipeline using DeepSeek-R1 across roughly 150 examples.
- The full-text strategy won 65.5% of head-to-head comparisons versus 12.2% for QA-only tuning, isolating training input coverage as the dominant factor for small-LLM QA performance.

TECHNICAL SKILLS

LANGUAGES

Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, Bash/Shell Scripting

AI-ASSISTED DEVELOPMENT

Claude Code, GitHub Copilot, OpenAI Codex, Notion AI

ML & AI

PyTorch, HuggingFace Transformers, LangChain, LangSmith, LoRA/PEFT, RAG Pipelines, FAISS, ChromaDB, MLflow, Weights & Biases

BACKEND & WEB

Flask, Django, REST APIs, React, .NET, C#, Streamlit

CLOUD, DEVOPS & DATA

Azure (AzureML), AWS (EC2, S3, Lambda), Docker, GitHub Actions, HPC/SLURM, Git, PostgreSQL, MySQL, CosmosDB, Oracle SQL